

STA326: Data Science Practice

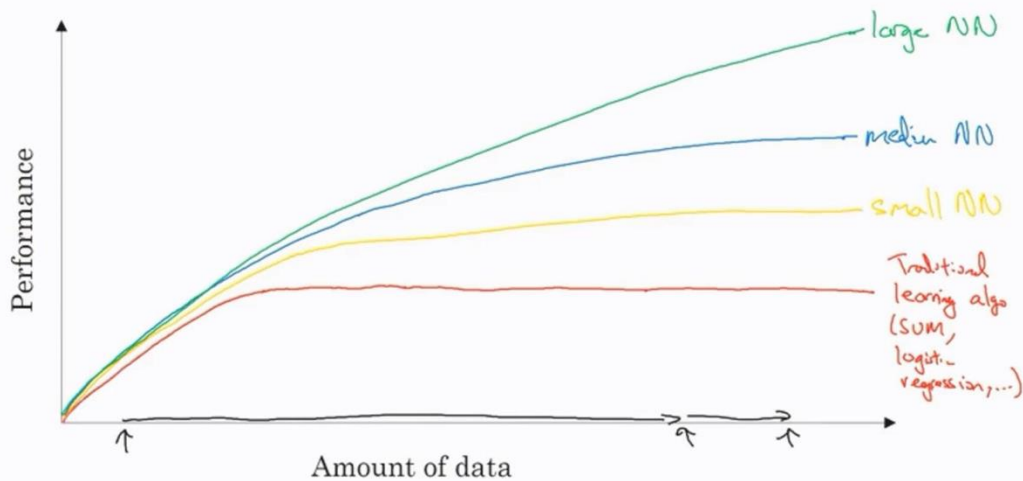
Wei Hongxin
2026 Spring

Quiz Time



Scale drives deep learning progress

网易云课堂



1. 下面哪种情况最有可能产生过拟合问题？ [单选题] *

模型参数量小，数据量小

模型参数量小，数据量大

模型参数量大，数据量小(正确答案)

模型参数量大，数据量大

2. 给定一个线性分类器 $O=XW+b$ ，已知单个输入特征 X 的维度为 $(1,5)$ 且包含3个标签类别，那么 weight 和 bias 的维度分别是： [单选题] *

$(3,5), (1,3)$

$(5,3), (1,3)$ (正确答案)

$(3,5), (1,5)$

$(3, 5), (1,5)$

3. 在线性回归中，我们推导MSE loss时所假设的真实数据 t 的分布是： [单选题] *

均匀分布

泊松分布

高斯分布(正确答案)

拉普拉斯分布

4. 我们通常采用哪个数据集用于模型选择和超参调优？ [单选题] *

训练集

验证集(正确答案)

测试集

以上都不是

5. 在机器学习中，交叉熵损失（CE Loss）、熵（Entropy）和 KL 散度（KL Divergence）之间存在以下关系。假设 P 是真实分布， Q 是模型预测的分布，下列哪个公式正确表示交叉熵损失与熵和 KL 散度的关系？

[单选题] *

$CE(P,Q) = Entropy(P) + KL(P || Q)$ (正确答案)

$CE(P,Q) = Entropy(P) - KL(P || Q)$

$CE(P,Q) = Entropy(Q) + KL(P || Q)$

$CE(P,Q) = Entropy(P) + KL(Q || P)$

A comparison of linear models

线性模型	激活函数	损失函数	优化方法
线性回归	-	$(y - \mathbf{w}^\top \mathbf{x})^2$	最小二乘、梯度下降
Logistic 回归	$\sigma(\mathbf{w}^\top \mathbf{x})$	$\mathbf{y} \log \sigma(\mathbf{w}^\top \mathbf{x})$	梯度下降
Softmax 回归	$\text{softmax}(\mathbf{W}^\top \mathbf{x})$	$\mathbf{y} \log \text{softmax}(\mathbf{W}^\top \mathbf{x})$	梯度下降
感知器	$\text{sgn}(\mathbf{w}^\top \mathbf{x})$	$\max(0, -y\mathbf{w}^\top \mathbf{x})$	随机梯度下降
支持向量机	$\text{sgn}(\mathbf{w}^\top \mathbf{x})$	$\max(0, 1 - y\mathbf{w}^\top \mathbf{x})$	二次规划、SMO 等

Deep Neural Networks

- Hidden layers
- Nonlinear – Activation Functions
- Gradient Vanishing and Exploding

Recall: Linear models

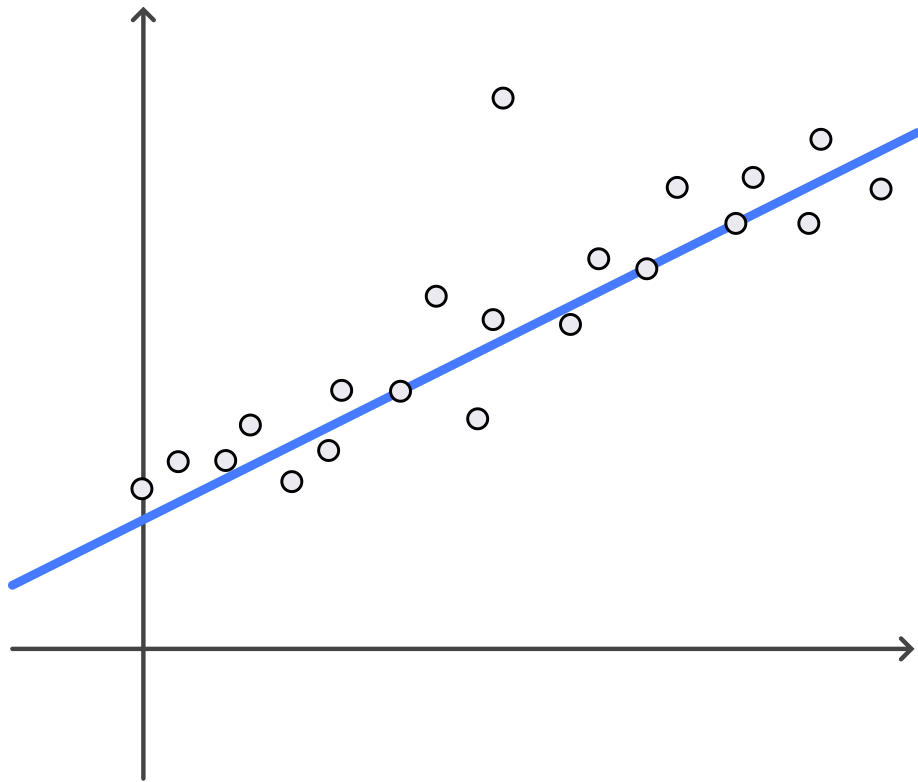
$$\mathbf{O} = \sigma(\mathbf{XW} + \mathbf{b}) \quad \sigma: \text{Softmax function}$$



A single affine transformation

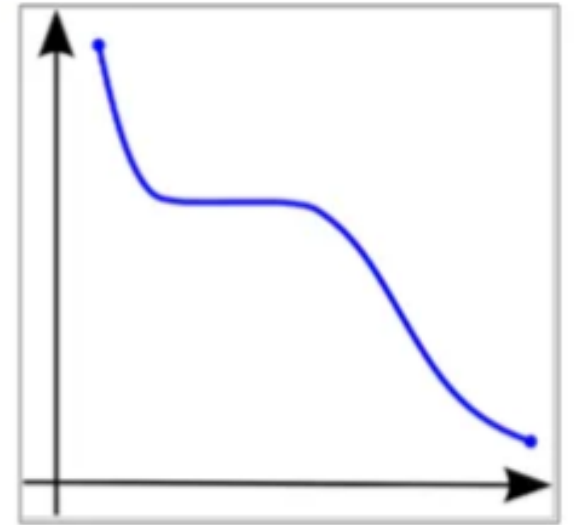
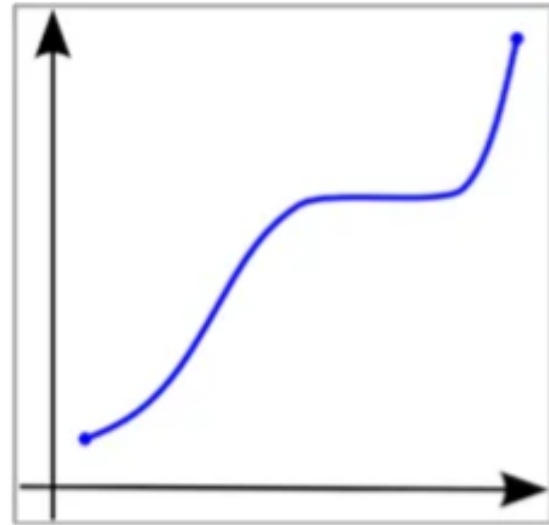
Is linear model sufficient for any predictions?

Is linear model sufficient?



Linearity

What is monotonicity :



Monotonicity (Weaker)

Example: Repay probs VS Income

Is it more possible for people with higher incomes to repay the loans?

Linear?

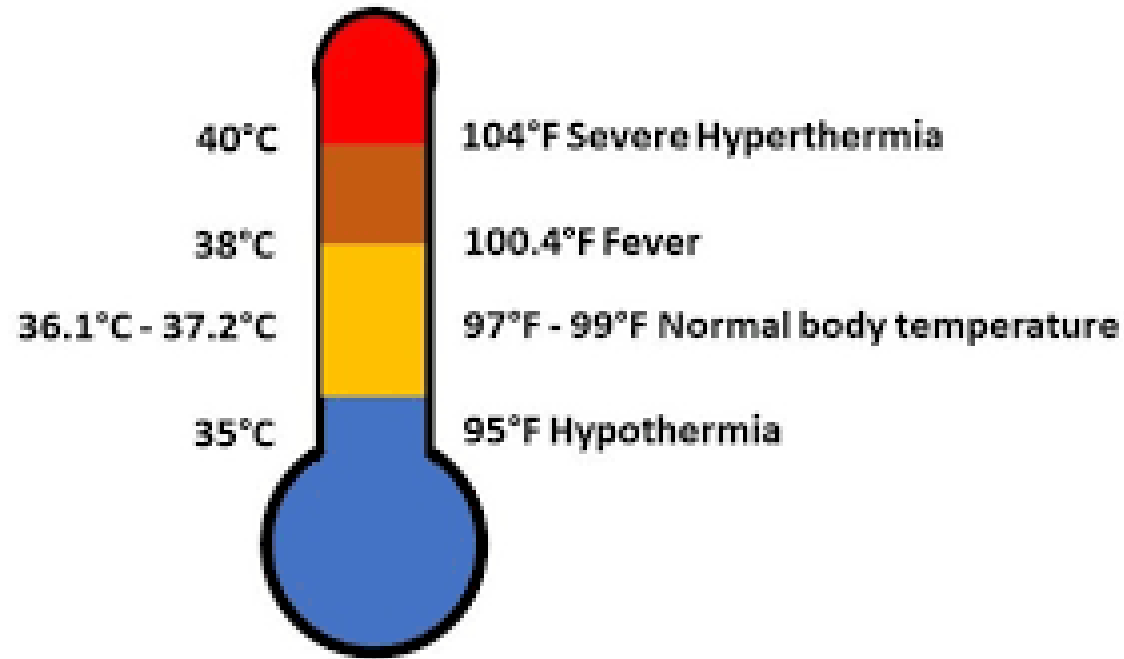
If not, how to make it linear?

Add a Log transformation on x



Fixed Basis function

Example of Violating the Monotonicity

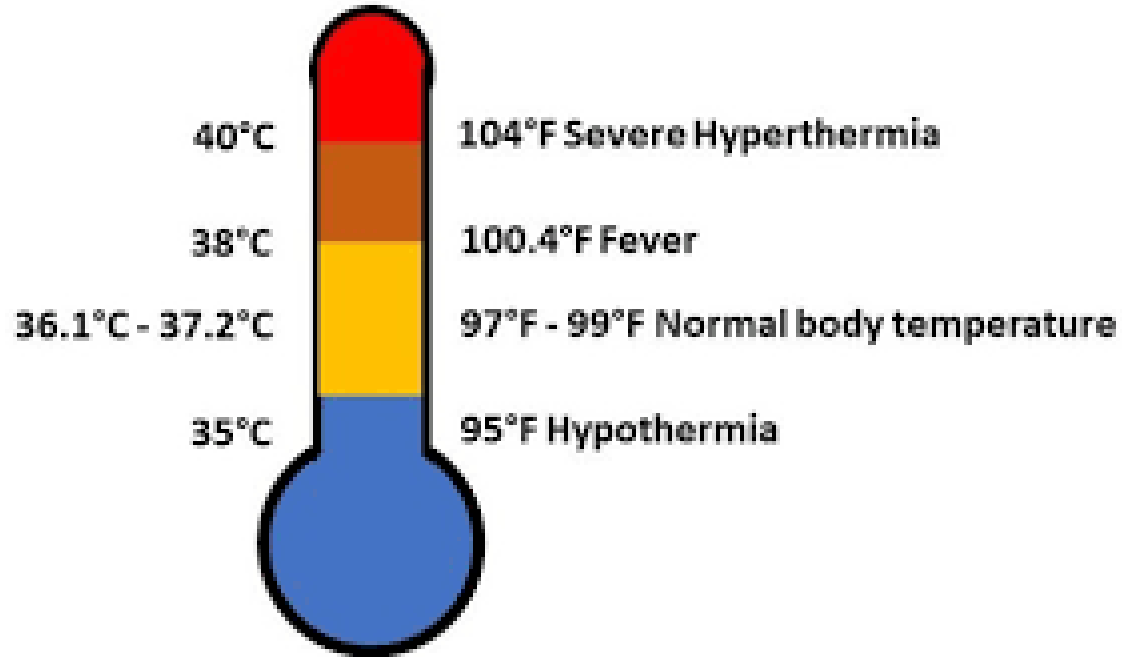


Predicting health based on body temperature

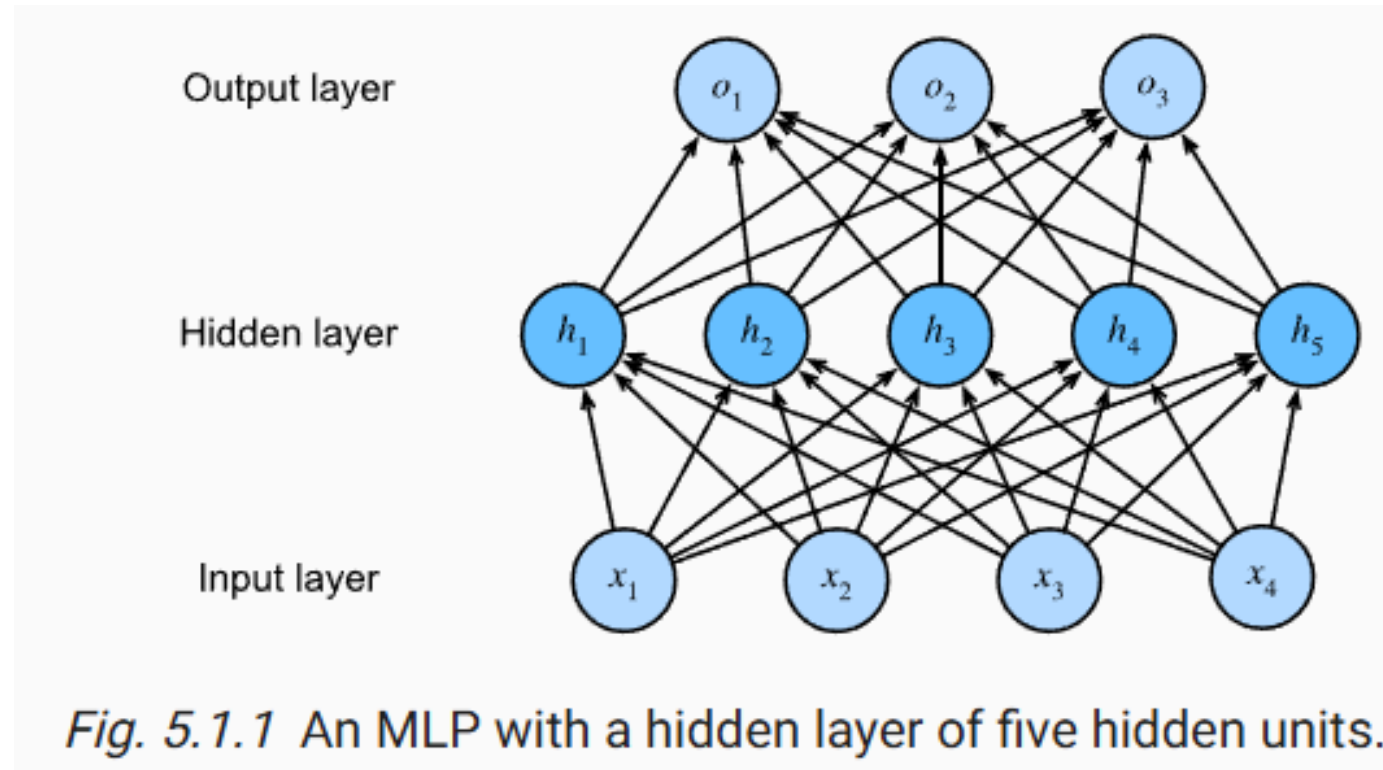
Example of image classification



Why fixed basis function is not enough?



Learning the feature with hidden layers



Learning the feature with hidden layers

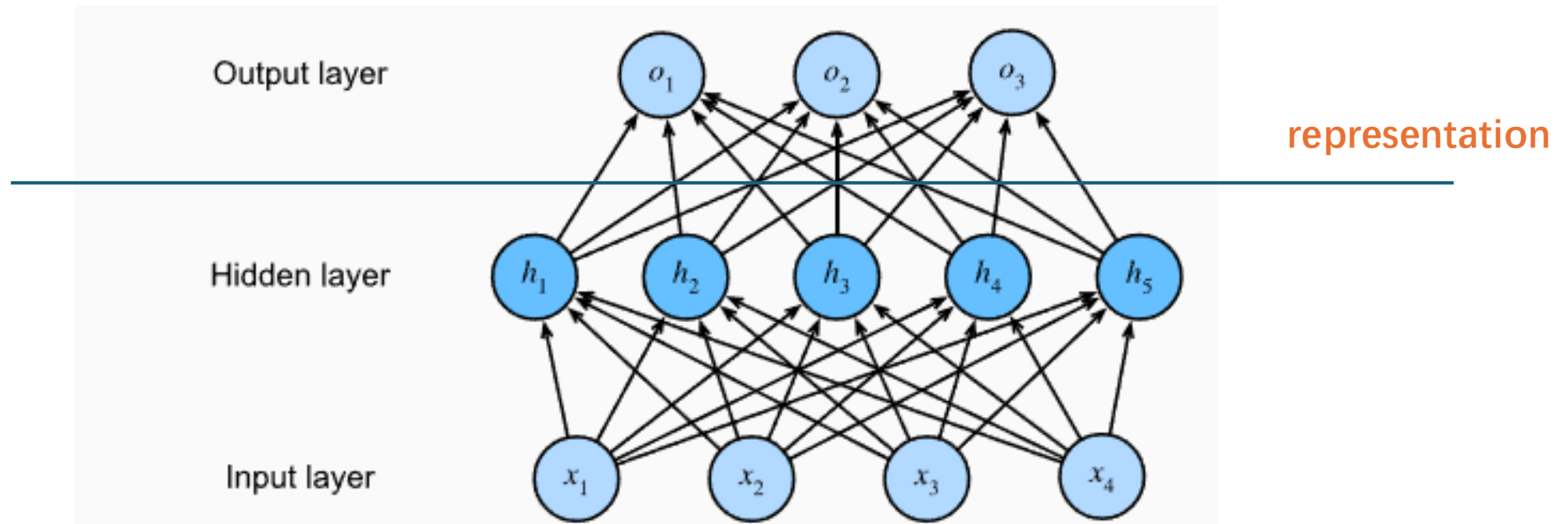


Fig. 5.1.1 An MLP with a hidden layer of five hidden units.

Multilayer Perceptron

Is MLP nonlinear?

$$\begin{aligned}\mathbf{H} &= \mathbf{XW}^{(1)} + \mathbf{b}^{(1)}, \\ \mathbf{O} &= \mathbf{HW}^{(2)} + \mathbf{b}^{(2)}.\end{aligned}$$

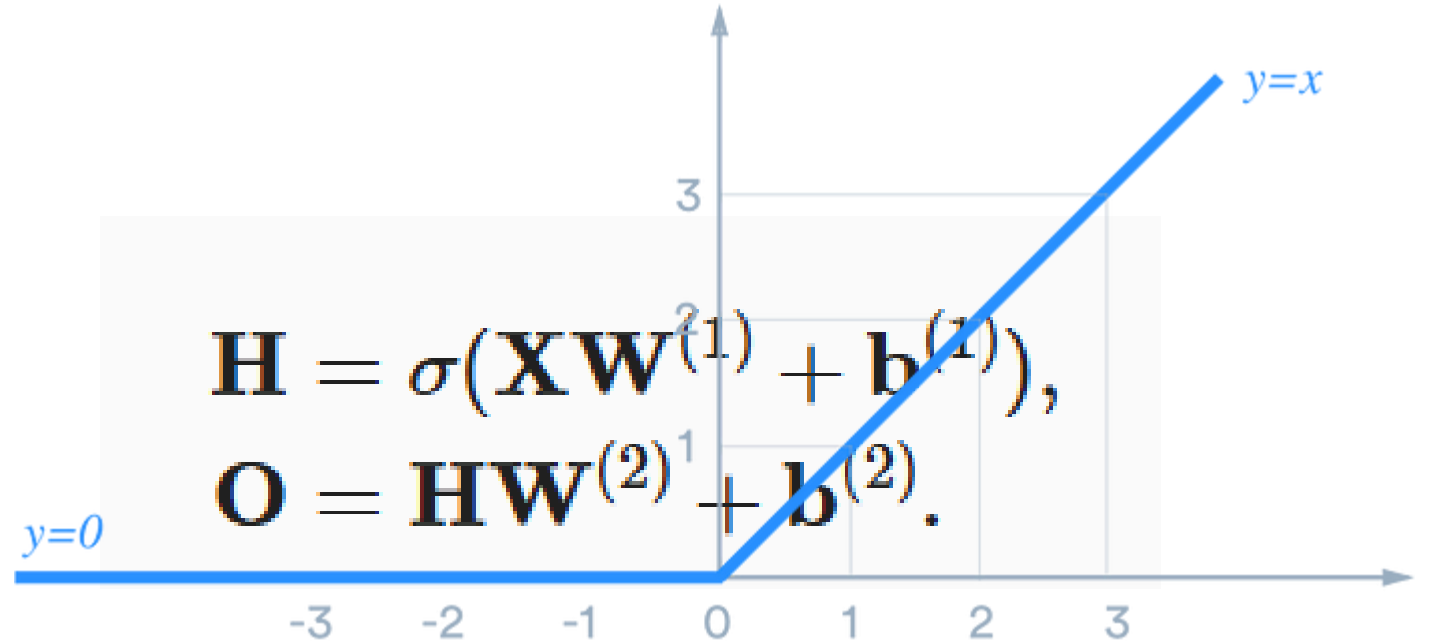
Hidden Layer

Output Layer

$$\mathbf{O} = (\mathbf{XW}^{(1)} + \mathbf{b}^{(1)})\mathbf{W}^{(2)} + \mathbf{b}^{(2)} = \mathbf{XW}^{(1)}\mathbf{W}^{(2)} + \mathbf{b}^{(1)}\mathbf{W}^{(2)} + \mathbf{b}^{(2)} = \mathbf{XW} + \mathbf{b}.$$

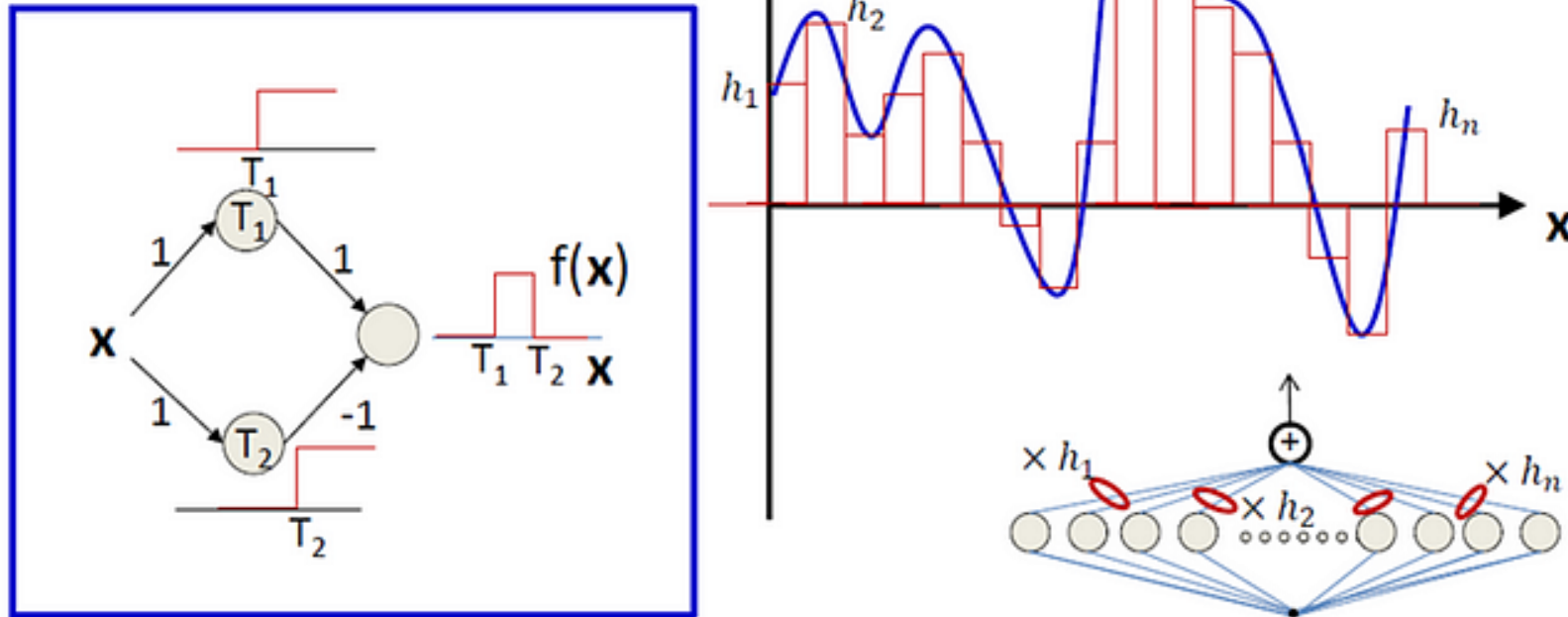
Equivalent to a single-layer model

We need nonlinear!



ReLU: $\sigma(x) = \max(0, x)$

Universal approximation theorem



Cybenko, G. (1989). Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals and Systems*, 2(4), 303–314.

定理 4.1 – 通用近似定理 (Universal Approximation Theorem)

[Cybenko, 1989, Hornik et al., 1989]: 令 $\varphi(\cdot)$ 是一个非常数、有界、单调递增的连续函数, $\mathcal{I}_d$ 是一个 d 维的单位超立方体 $[0, 1]^d$, $C(\mathcal{I}_d)$ 是定义在 $\mathcal{I}_d$ 上的连续函数集合。对于任何一个函数 $f \in C(\mathcal{I}_d)$, 存在一个整数 m , 和一组实数 $v_i, b_i \in \mathbb{R}$ 以及实数向量 $\mathbf{w}_i \in \mathbb{R}^d$, $i = 1, \dots, m$, 以至于我们可以定义函数

$$F(\mathbf{x}) = \sum_{i=1}^m v_i \varphi(\mathbf{w}_i^T \mathbf{x} + b_i), \quad (4.33)$$

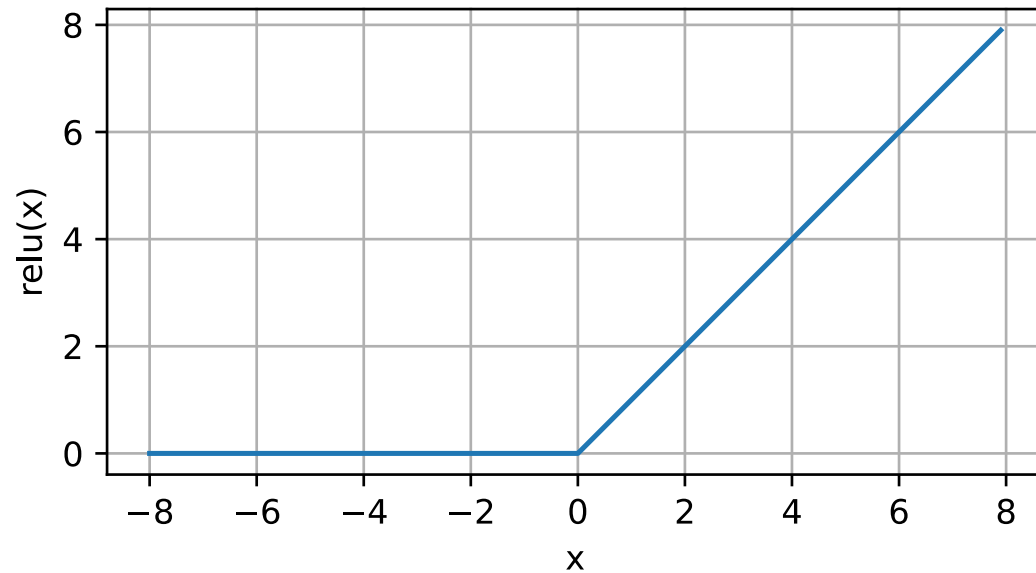
作为函数 f 的近似实现, 即

$$|F(\mathbf{x}) - f(\mathbf{x})| < \epsilon, \forall \mathbf{x} \in \mathcal{I}_d. \quad (4.34)$$

其中 $\epsilon > 0$ 是一个很小的正数。

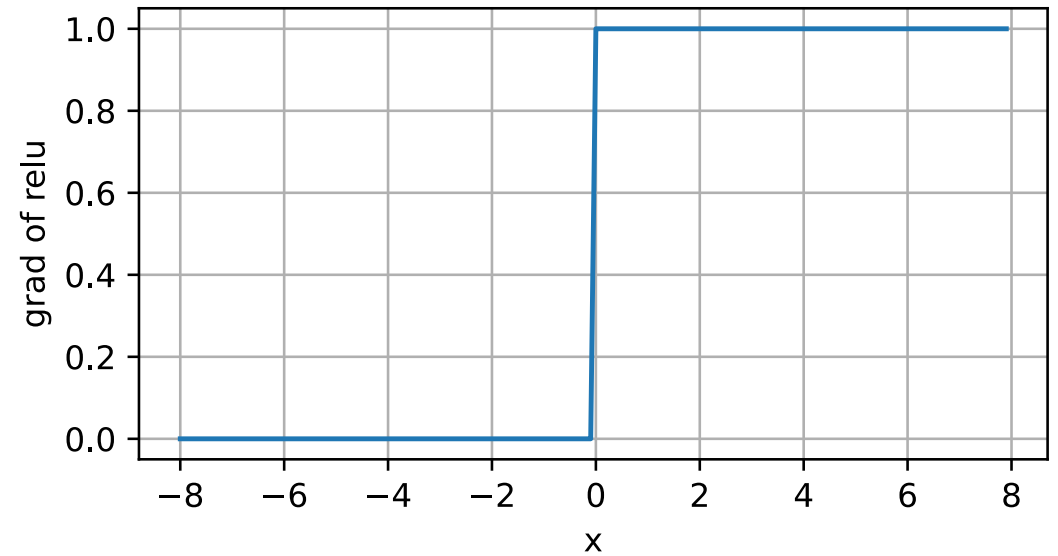
根据通用近似定理, 对于具有线性输出层和至少一个使用“挤压”性质的激活函数的隐藏层组成的前馈神经网络, 只要其隐藏层神经元的数量足够, 它可以以任意的精度来近似任何从一个定义在实数空间中的有界闭集函数。

Activation Functions



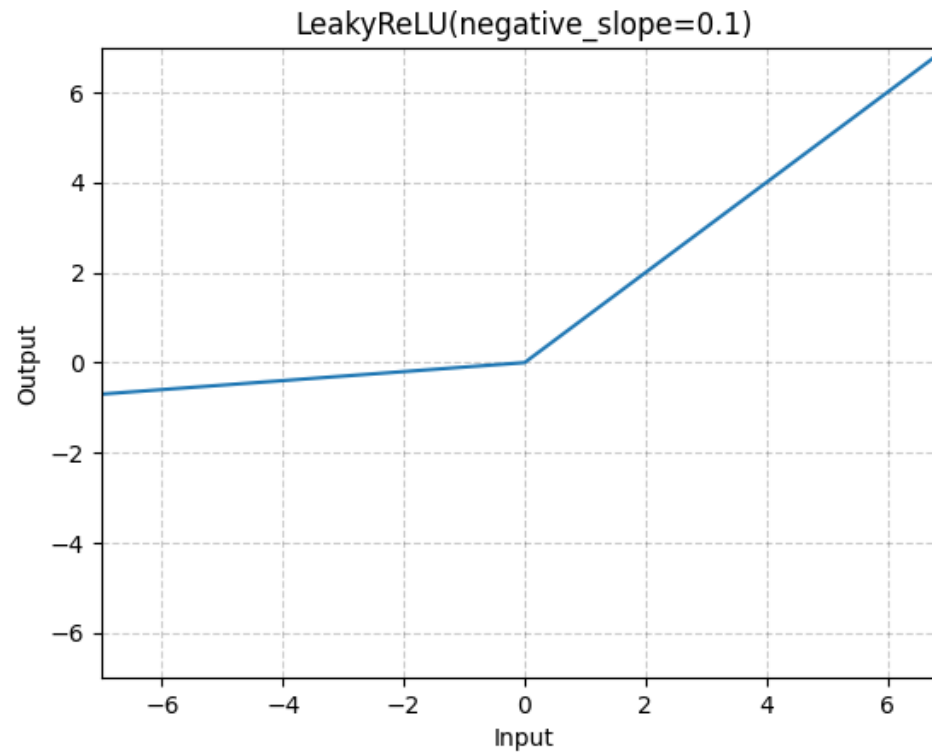
$$\text{ReLU}(x) = \max(x, 0).$$

rectified linear unit (ReLU)



Gradient of ReLU

Variants of ReLU

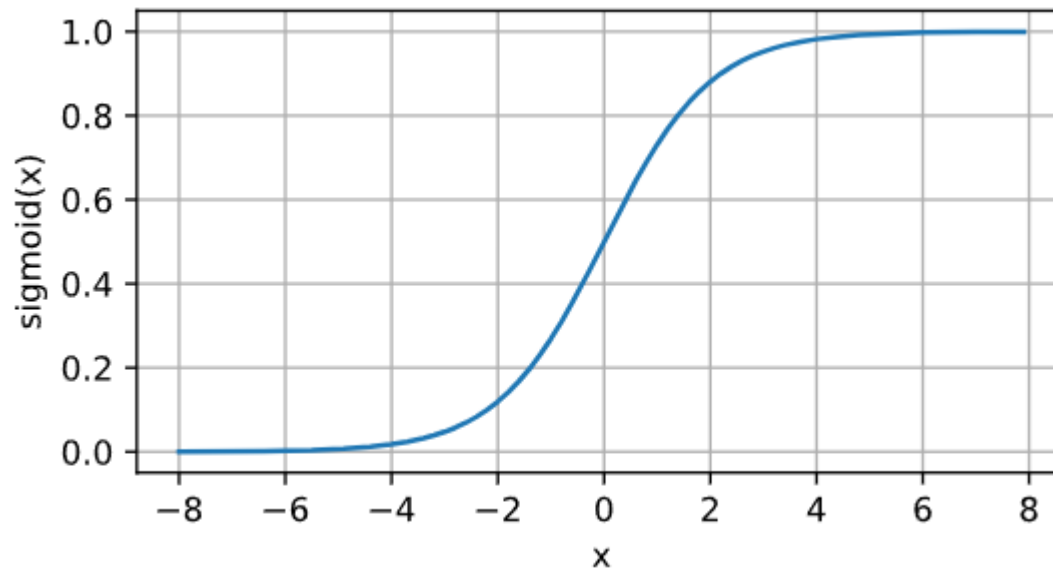


$$\text{pReLU}(x) = \max(0, x) + \alpha \min(0, x).$$

LeakyReLU

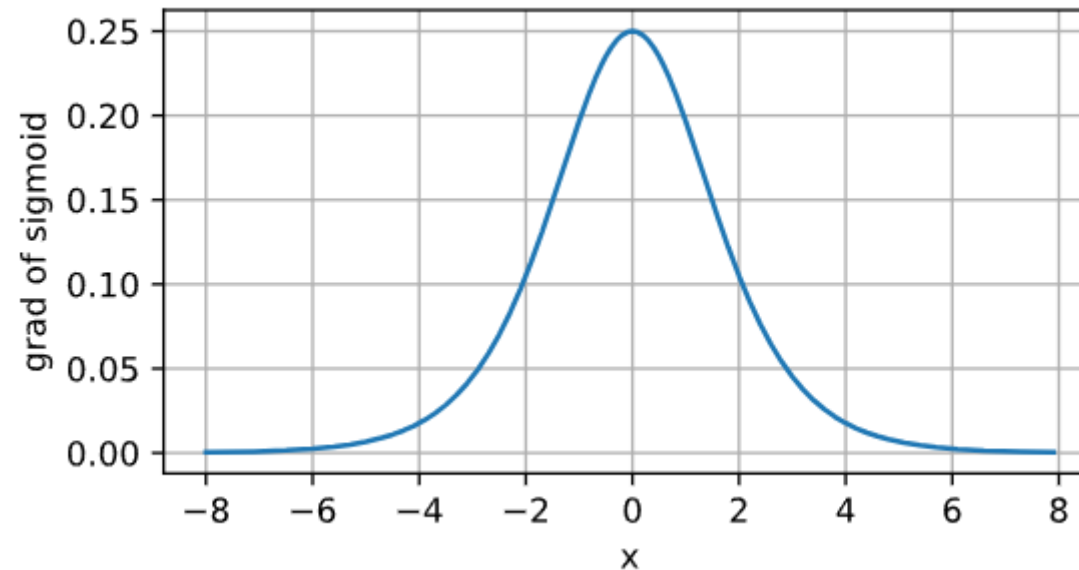
Sigmoid

$$\text{sigmoid}(x) = \frac{1}{1 + \exp(-x)}.$$



Gradient of Sigmoid

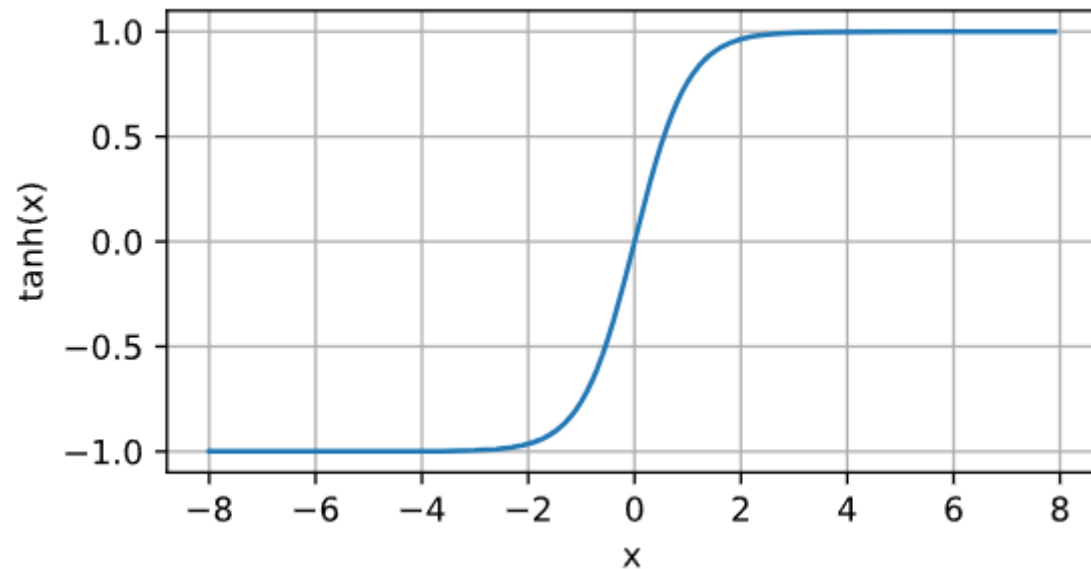
$$\frac{d}{dx} \text{sigmoid}(x) = \frac{\exp(-x)}{(1 + \exp(-x))^2} = \text{sigmoid}(x) (1 - \text{sigmoid}(x)).$$



LeCun, Y., Bottou, L., Orr, G., & Muller, K.-R. (1998). Efficient backprop. *Neural Networks: Tricks of the Trade*. Springer.

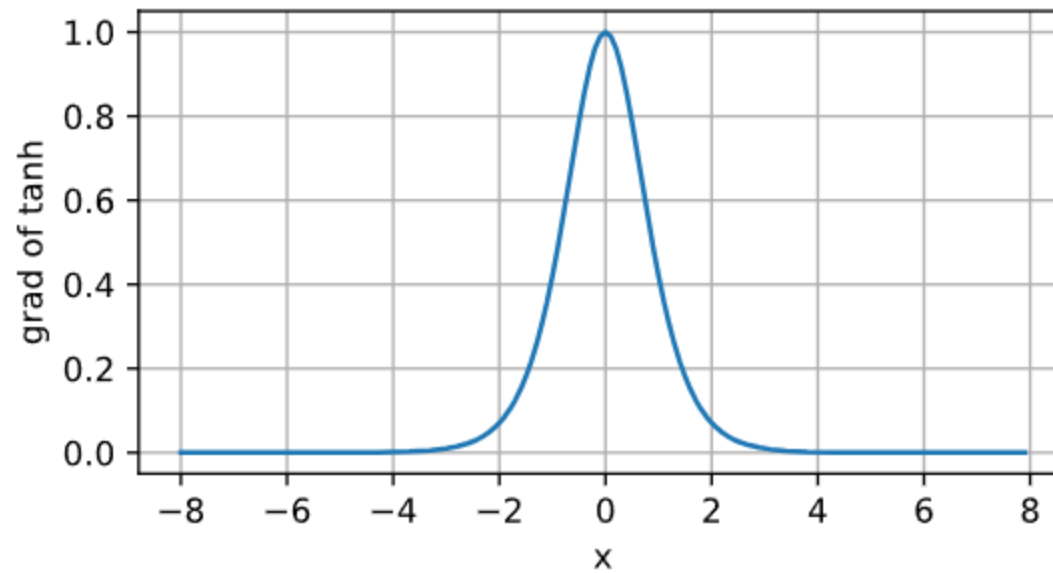
Tanh

$$\tanh(x) = \frac{1 - \exp(-2x)}{1 + \exp(-2x)}.$$



Gradient of Sigmoid

$$\frac{d}{dx} \tanh(x) = 1 - \tanh^2(x).$$



Comparison of activations

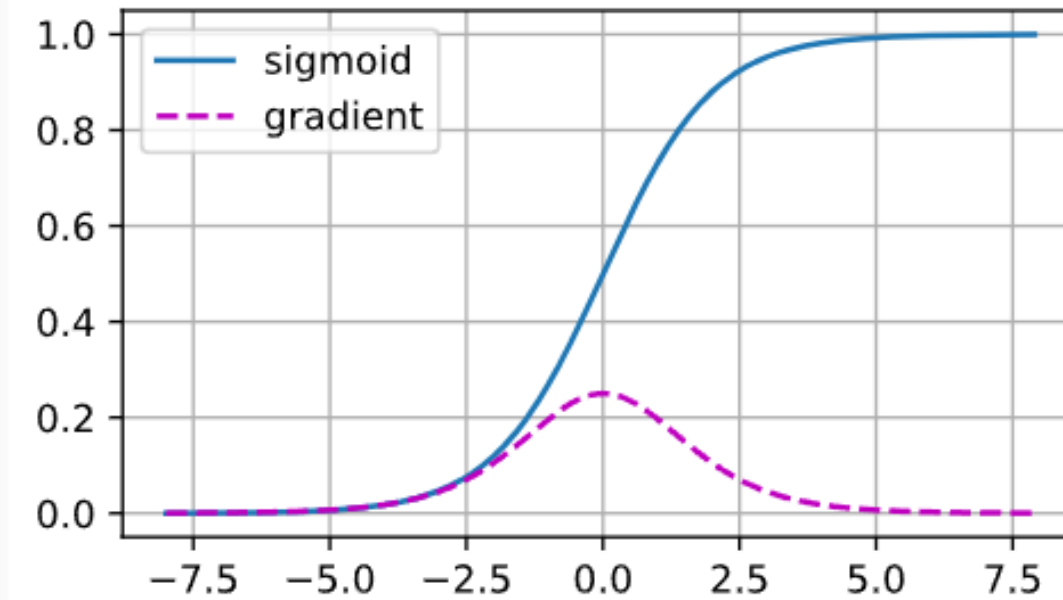
ReLU: 简单、稀疏、无梯度消失（但有死神经元）。

Sigmoid: 平滑、概率输出（但梯度消失）。

Tanh: 对称、范围广（但优化仍难）。

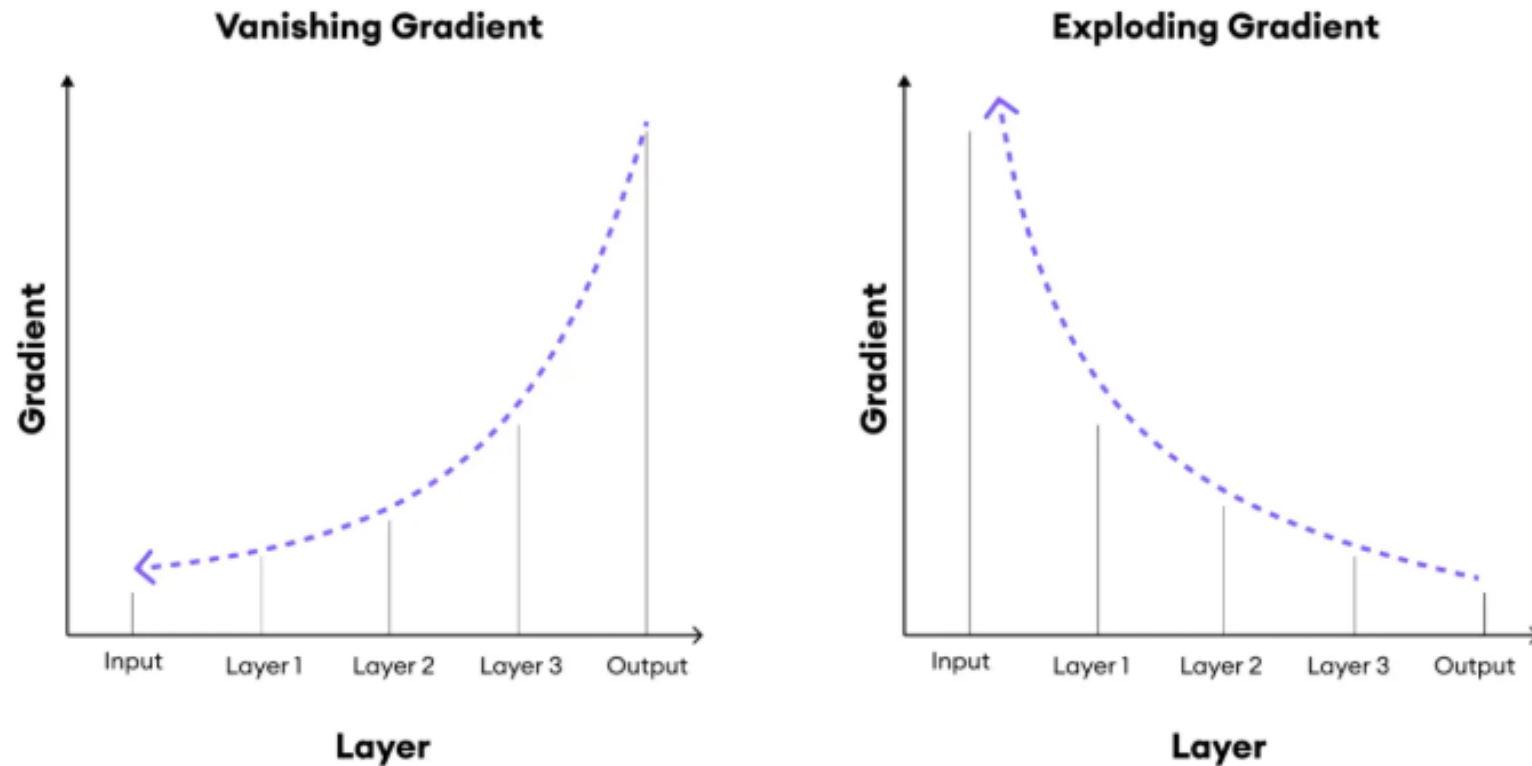
How to choose?

Gradient Vanishing



$$\partial_{\mathbf{w}^{(l)}} \mathbf{o} = \underbrace{\partial_{\mathbf{h}^{(L-1)}} \mathbf{h}^{(L)}}_{\mathbf{M}^{(L)} \stackrel{\text{def}}{=}} \cdots \underbrace{\partial_{\mathbf{h}^{(l)}} \mathbf{h}^{(l+1)}}_{\mathbf{M}^{(l+1)} \stackrel{\text{def}}{=}} \underbrace{\partial_{\mathbf{w}^{(l)}} \mathbf{h}^{(l)}}_{\mathbf{v}^{(l)} \stackrel{\text{def}}{=}}.$$

Gradient Exploding



Solutions: Gradient clipping, Xavier Initialization

Conclusion

- Why do we need hidden layers?
- Why do we need activation functions?
- What's the weakness of Sigmoid?

Extension Question?

- What will happen if we set the initial weights as 0?
- Why we choose deeper networks instead of wider?

Problem Description:

Design an experiment to show that a two-layer ReLU network can approximate any function with an example. Define a custom function and fit it using a ReLU-based neural network.

Requirements:

1. Generate training and test sets by sampling from the defined function.
2. Train the neural network using the training set.
3. Validate the fitting performance using the test set.

Survey



Next course

- **Optimization in DNN**

Chapter 12 in 《dive into deep learning》

Chapter 7 in 《Deep leanning》